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(54) **MULTIPHASE-TO-SINGLE-PHASE AUTOMATIC TRANSFER SWITCH**

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(52) **U.S. Cl.**
CPC ***H02J 9/068*** (2020.01); ***H02J 9/061*** (2013.01)

(58) **Field of Classification Search**
CPC H02J 9/068; H02J 9/061
USPC 307/23
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,761,563 A * 8/1988 Ross H02J 9/066 307/64
7,064,458 B2 6/2006 Wittner
7,762,786 B2 7/2010 Taylor et al.
7,928,604 B2 4/2011 Page
9,467,006 B2 10/2016 Dickerson et al.
9,996,128 B2 6/2018 Chapel et al.
10,601,246 B2 3/2020 Ding et al.
10,742,030 B2 8/2020 Sharifipour
10,840,735 B1 11/2020 Cooper
10,928,878 B2 2/2021 Morales et al.
10,985,603 B2 4/2021 Tomassi
11,211,816 B1 12/2021 Bose et al.
(Continued)

FOREIGN PATENT DOCUMENTS

EP 2672603 A1 12/2013
EP 2665346 B1 7/2019
(Continued)

OTHER PUBLICATIONS

Extended European Search Report from corresponding European Application No. 24218991.8 dated May 19, 2025.

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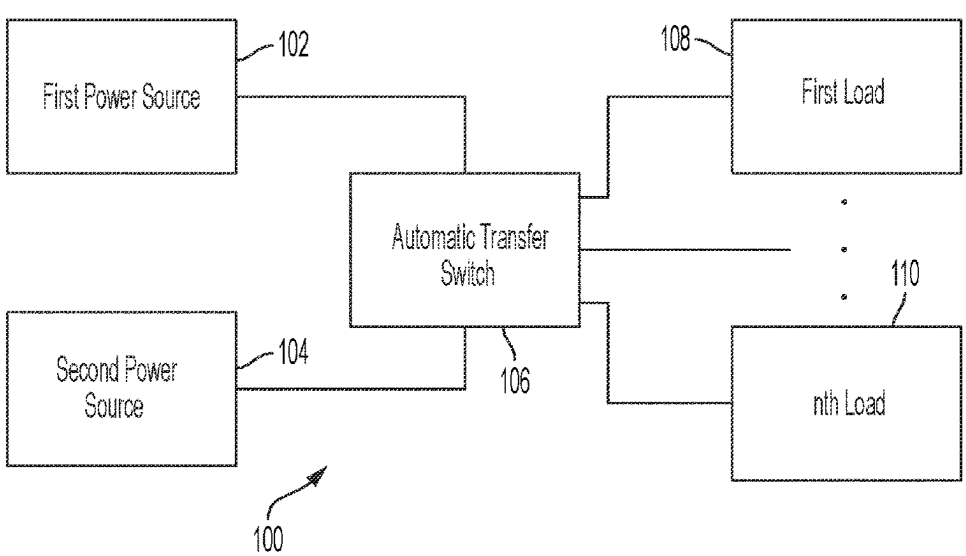
(57) **ABSTRACT**

Examples of the disclosure include an automatic transfer switching system comprising a first input to receive phases of first power from a first source, a second input to receive phases of second power from a second source, a receptacle, a phase-selection interface (PSI) configured to receive the first power and output a first selected phase and receive the second power and output a second selected phase, a switching device coupled to the receptacle and the PSI, and at least one controller configured to control the switching device to couple the first input to the receptacle via the PSI to provide the first selected phase when the switching device decouples the second input from the receptacle, and to control the switching device to couple the second input to the receptacle via the PSI to provide the second selected phase when the switching device decouples the first input from the receptacle.

20 Claims, 8 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

2005/0141154	A1	6/2005	Consadori et al.	
2012/0078428	A1	3/2012	Henderieckx	
2022/0247304	A1*	8/2022	Oudrhiri	G06F 1/30

FOREIGN PATENT DOCUMENTS

WO	2015094290	A1	6/2015
WO	2022162415	A1	8/2022

* cited by examiner

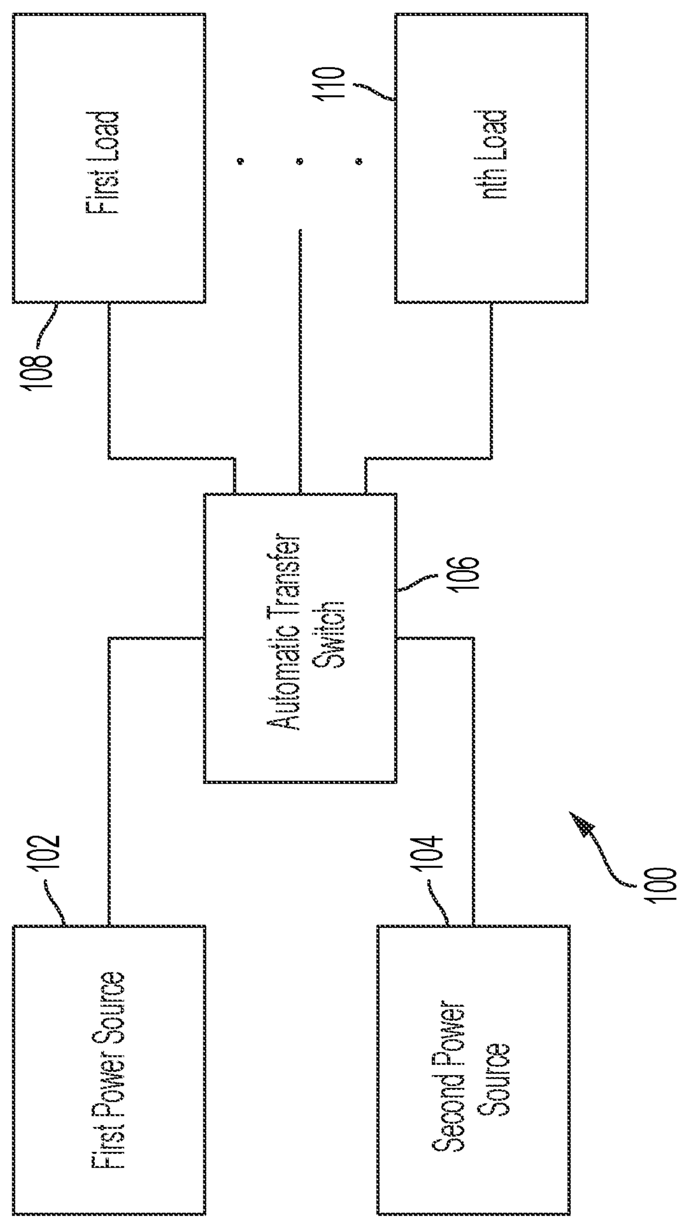

First Load
nth Load
110
108
Automatic Transfer Switch
106
First Power Source
Second Power Source
102
104
100

FIG. 1

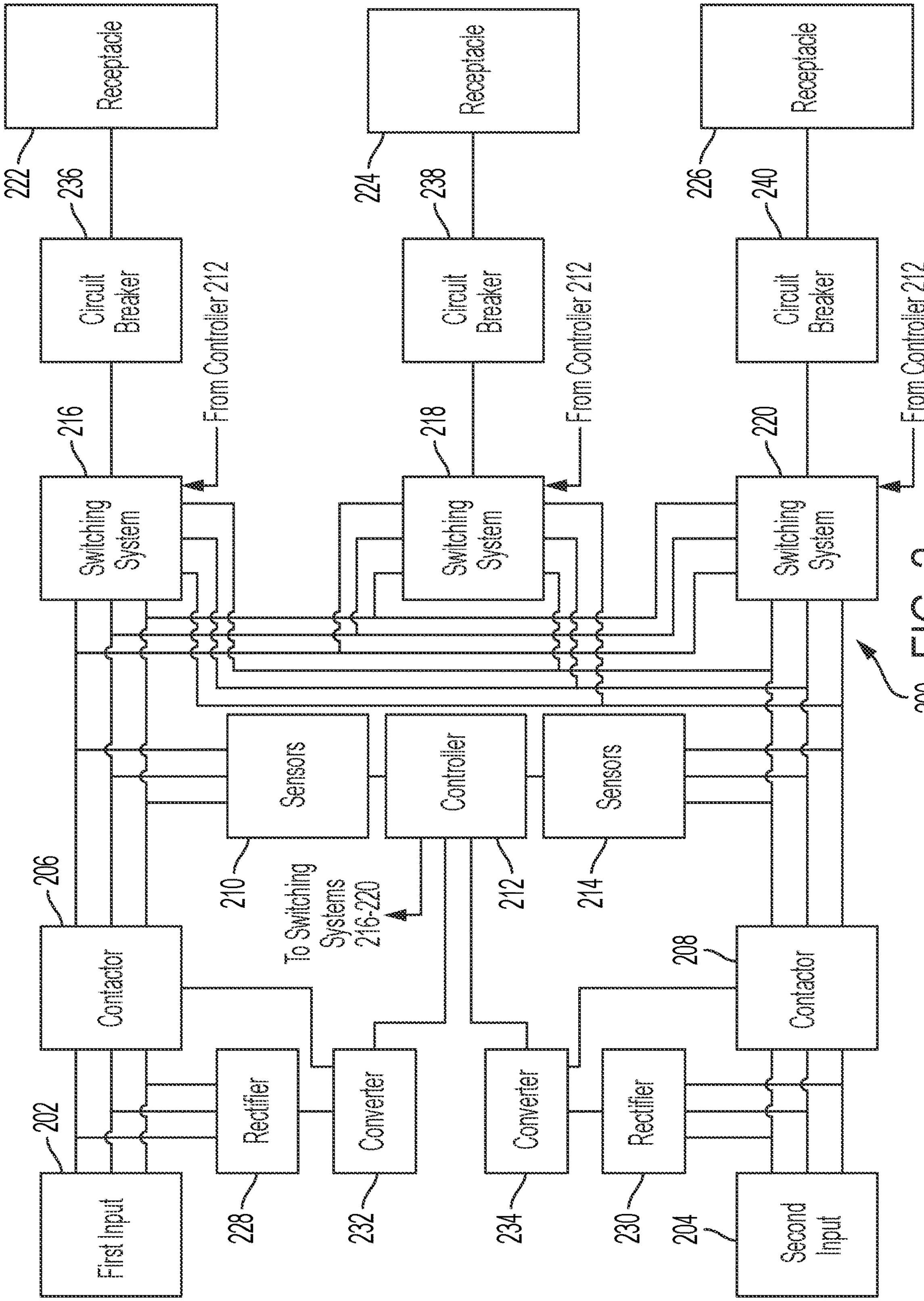

Receptacle
222
Receptacle
224
Receptacle
226
Circuit Breaker
236
Circuit Breaker
238
Circuit Breaker
240
Switching System
216
Switching System
218
Switching System
220
From Controller 212
From Controller 212
From Controller 212
Sensors
210
Controller
212
Sensors
214
To Switching Systems 216-220
Contactor
206
Contactor
208
Rectifier
228
Converter
232
Converter
234
Rectifier
230
First Input
202
Second Input
204
200

FIG. 2

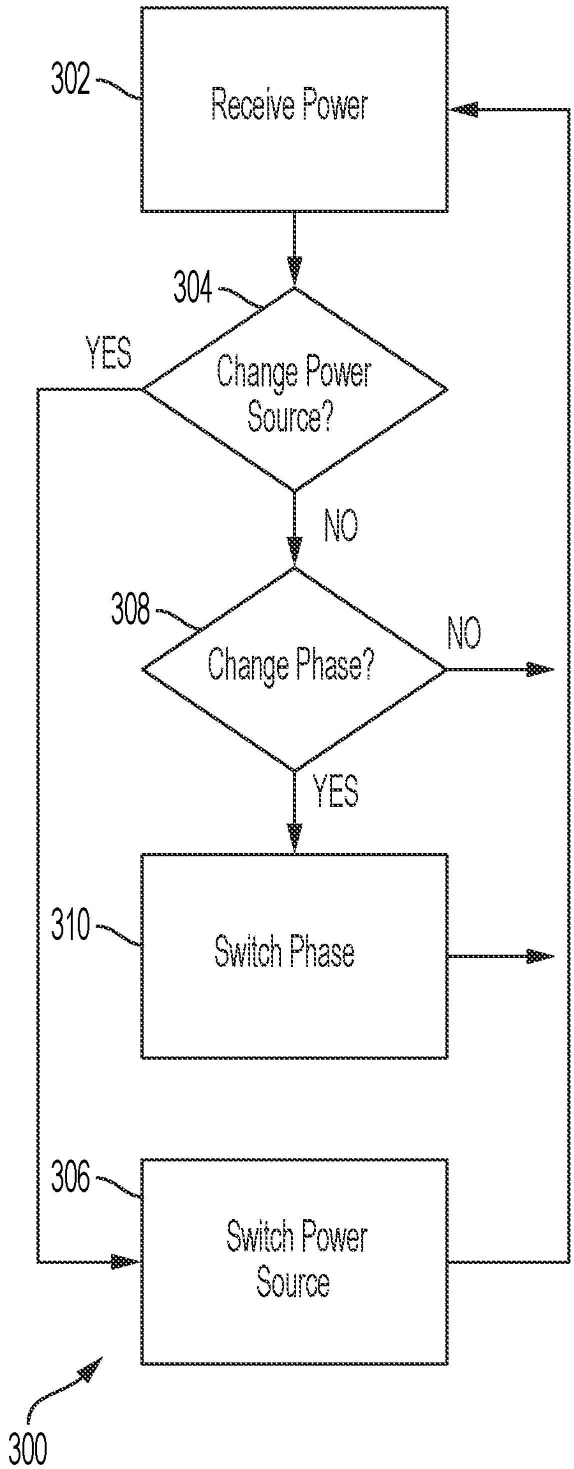

302
Receive Power
304
Change Power Source?
YES
NO
308
Change Phase?
NO
YES
310
Switch Phase
306
Switch Power Source
300

FIG. 3

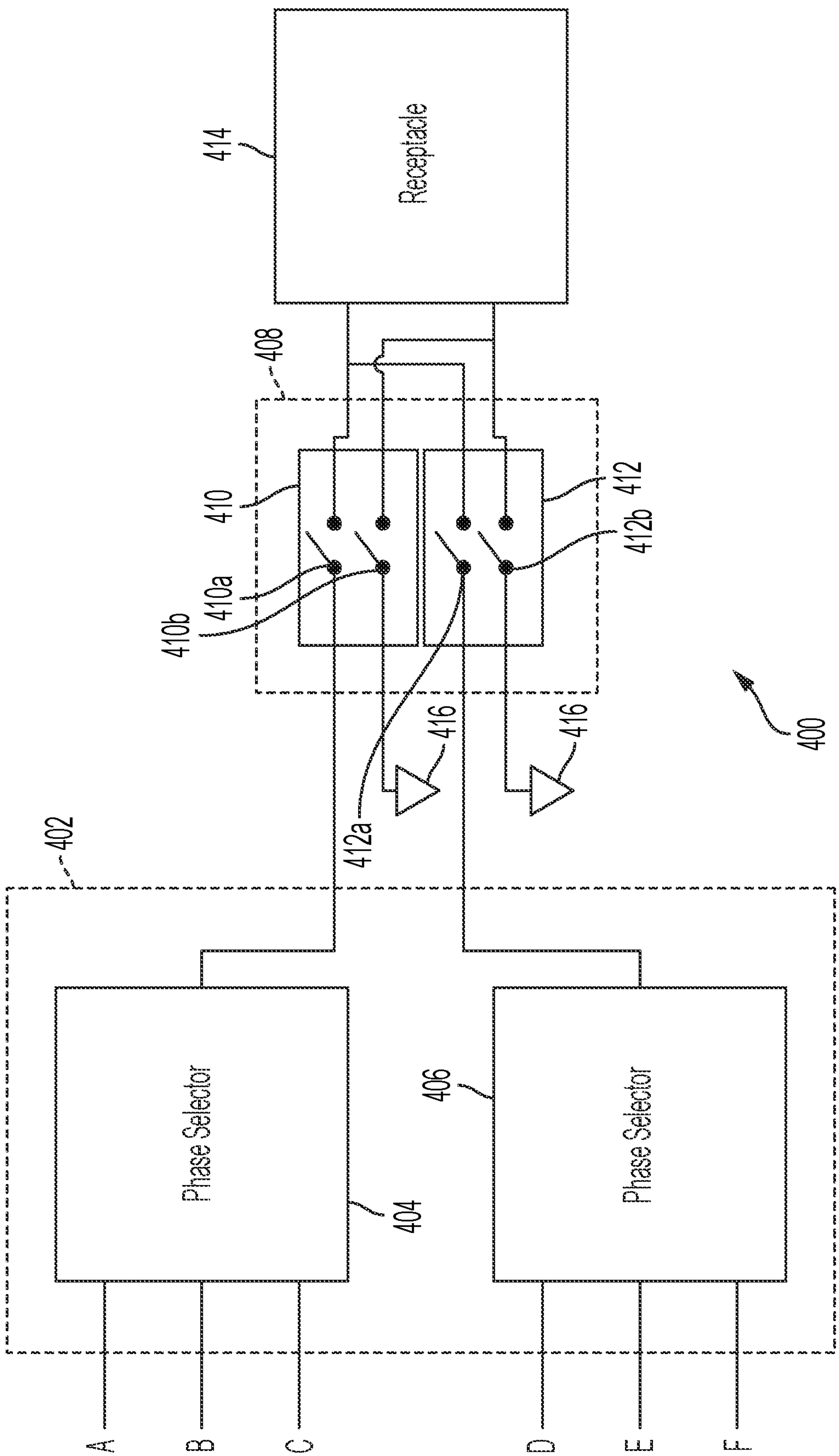

414
Receptacle
408
410
412
410a
410b
412a
412b
416
416
400
402
Phase Selector
404
406
Phase Selector
A
B
C
D
E
F

FIG. 4

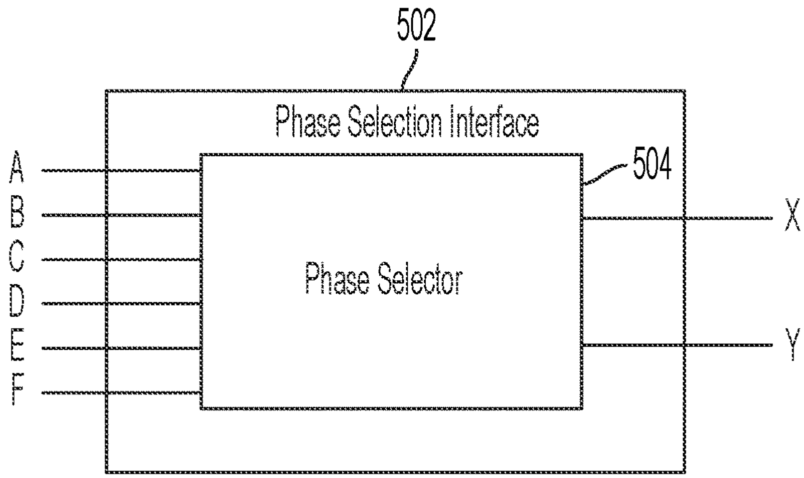

502
Phase Selection Interface
504
A
B
C
D
E
F
Phase Selector
X
Y

FIG. 5A

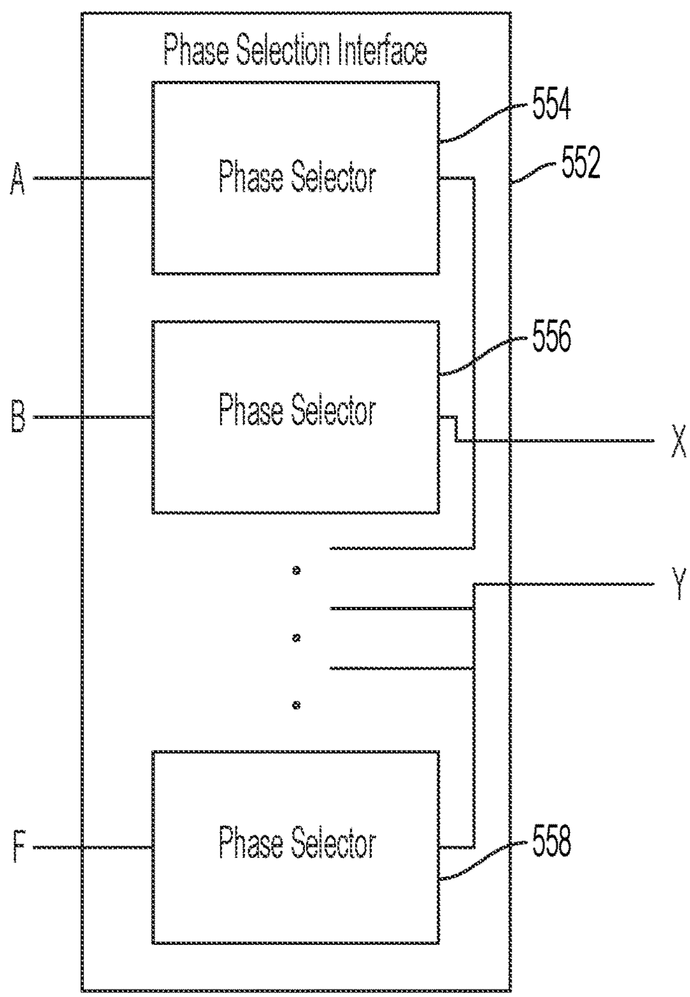

Phase Selection Interface
554
552
A
Phase Selector
556
B
Phase Selector
X
Y
F
Phase Selector
558

FIG. 5B

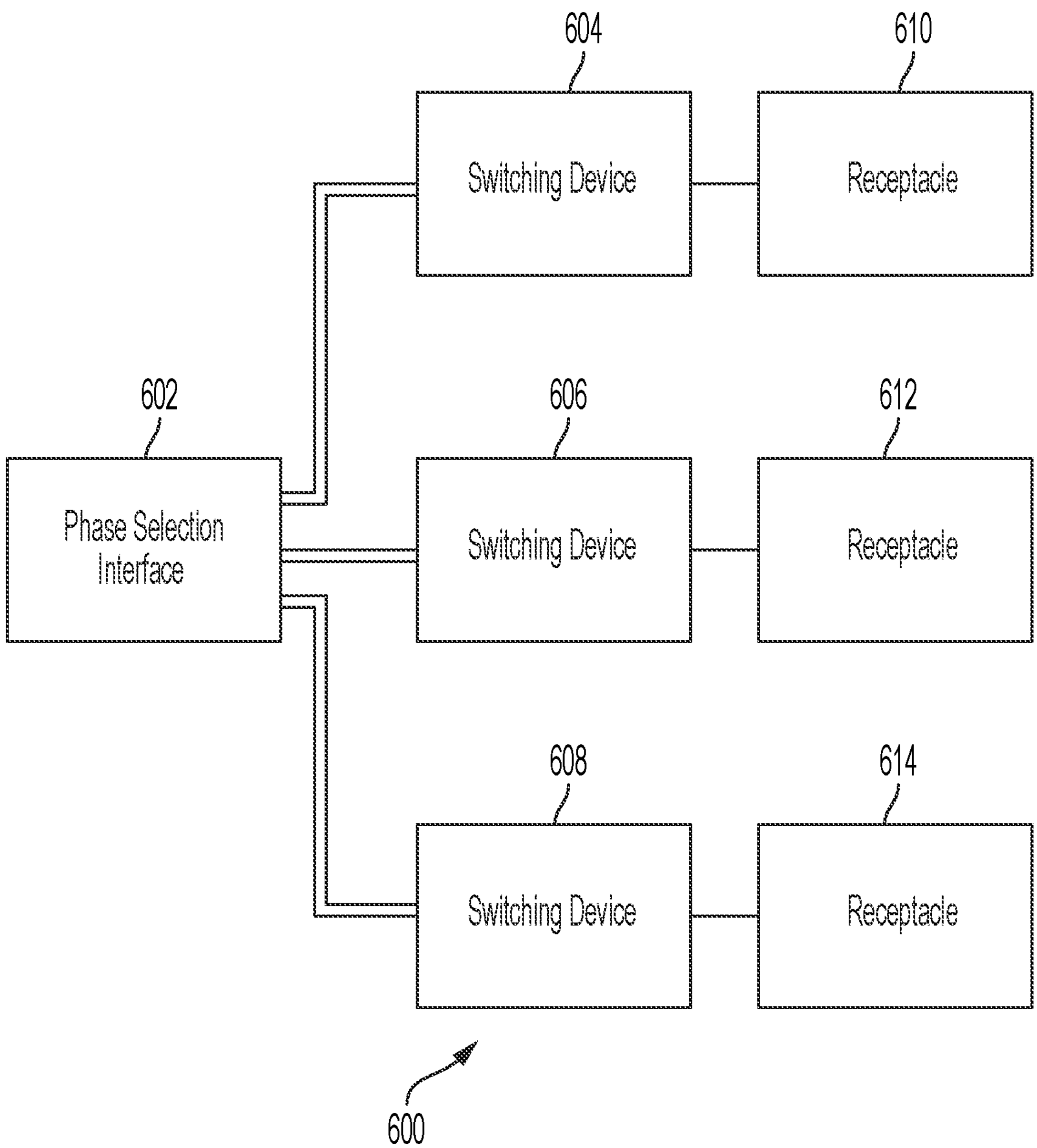

604
610
Switching Device
Receptacle
602
606
612
Phase Selection Interface
Switching Device
Receptacle
608
614
Switching Device
Receptacle
600

FIG. 6

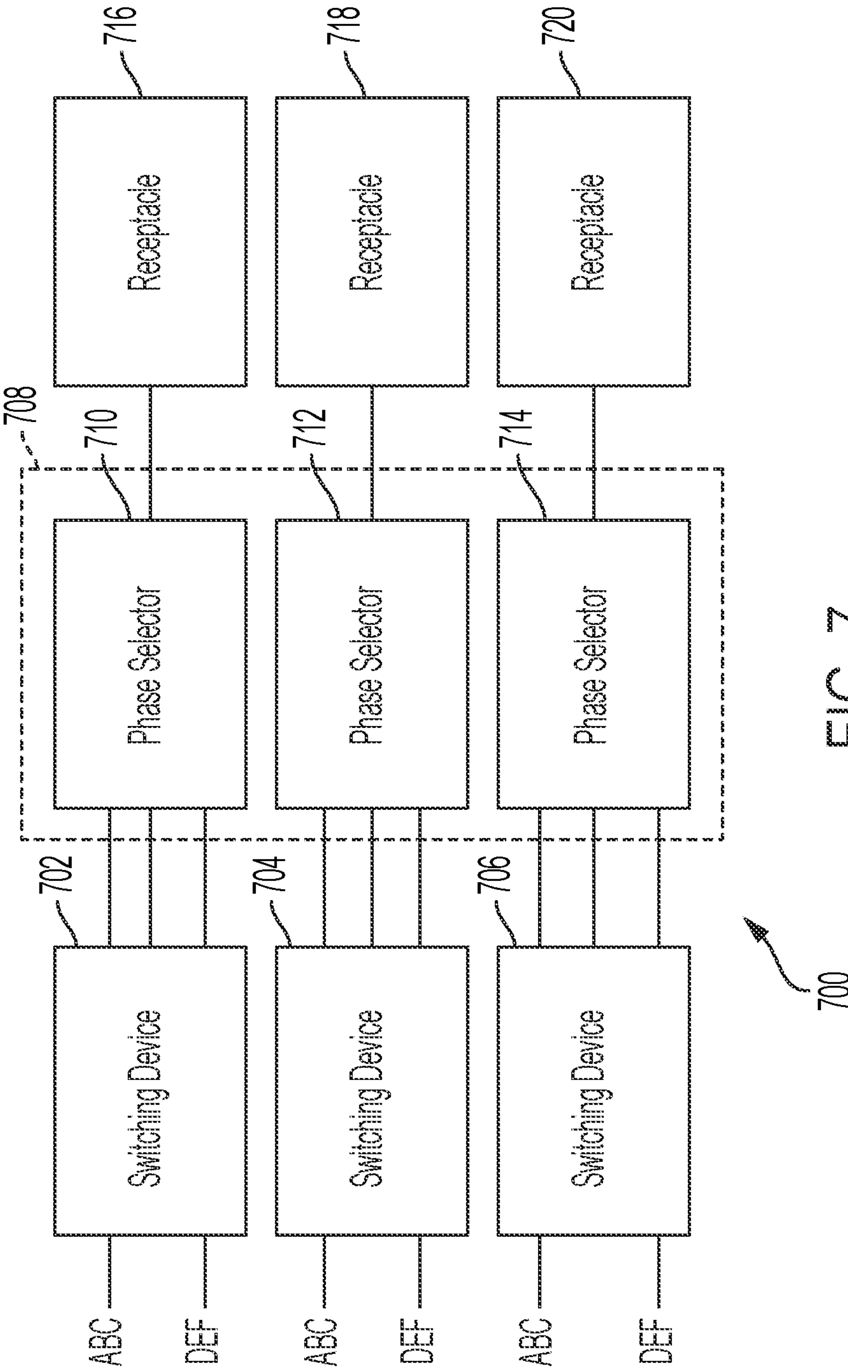

716
718
720
Receptacle
Receptacle
Receptacle
708
710
712
714
Phase Selector
Phase Selector
Phase Selector
702
704
706
Switching Device
Switching Device
Switching Device
ABC
DEF
ABC
DEF
ABC
DEF
700

FIG. 7

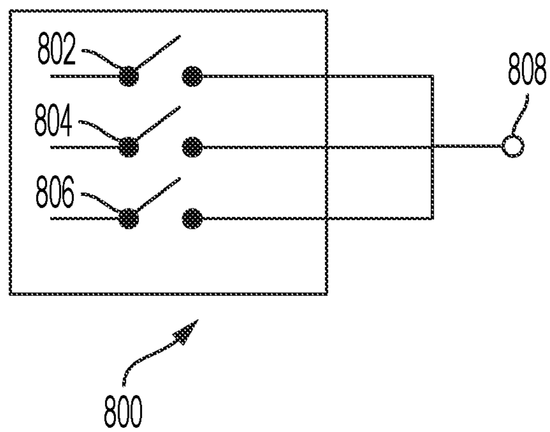


FIG. 8A

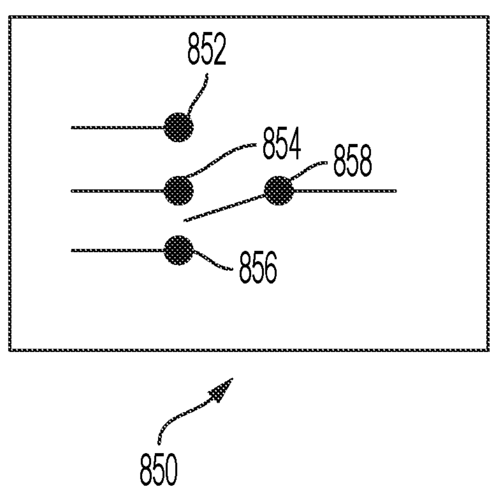


FIG. 8B

MULTIPHASE-TO-SINGLE-PHASE AUTOMATIC TRANSFER SWITCH

BACKGROUND

1. Field of the Disclosure

At least one example in accordance with the present disclosure relates generally to power-transfer systems.

2. Discussion of Related Art

Server racks in data centers may use large amounts of power. Some server racks may be connected to multiple main-power sources. If power becomes unavailable from one main-power source, the server rack may switch to drawing power from another main-power source.

SUMMARY

According to at least one aspect of the present disclosure, an automatic transfer switching (ATS) system is provided comprising a first input configured to receive multiple phases of first power from a first power source, a second input configured to receive multiple phases of second power from a second power source, a receptacle configured to provide power to one or more loads, at least one phase-selection interface configured to receive the multiple phases of first power from the first power source and selectively output a first selected phase of the first power chosen from among the multiple phases of the first power, and receive the multiple phases of second power from the second power source and selectively output a second selected phase of the second power chosen from among the multiple phases of the second power, at least one switching device coupled to the receptacle and to the at least one phase-selection interface, and at least one controller configured to control the at least one switching device to provide power to the receptacle, wherein controlling the at least one switching device to provide power to the receptacle includes controlling the at least one switching device to couple the first input to the receptacle via the at least one phase-selection interface to provide the first selected phase to the receptacle when the at least one switching device decouples the second input from the receptacle, and controlling the at least one switching device to couple the second input to the receptacle via the at least one phase-selection interface to provide the second selected phase to the receptacle when the at least one switching device decouples the first input from the receptacle.

In at least one example, the at least one switching device includes a first switch and a second switch, the first switch being configured to selectively couple the receptacle to the first input and the second switch being configured to selectively couple the receptacle to the second input. In at least one example, the first switch is in a closed state when the first switch couples the first input to the receptacle, the second switch is in a closed state when the second switch couples the second input to the receptacle, the first switch is in an open state when the first switch decouples the first input from the receptacle, and the second switch is in an open state when the second switch decouples the second input from the receptacle.

In at least one example, the at least one phase-selection interface includes a phase selection switch configured to selectively couple at least one phase input of the first input to the receptacle, the at least one phase input being configured to receive at least one phase of the multiple phases of first power from a power source. In at least one example, the at least one phase-selection interface includes a phase selection switch configured to selectively couple at least one phase input of the first input to the at least one switching device, the at least one phase input being configured to receive at least one phase of the multiple phases of first power from a power source. In at least one example, the at least one phase-selection interface includes a phase selection switch configured to selectively couple at least one phase input of the second input to the receptacle, the at least one phase input being configured to receive at least one phase of the multiple phases of second power from a power source.

In at least one example, the at least one phase-selection interface includes a phase selection switch configured to selectively couple at least one phase input of the second input to the at least one switching device, the at least one phase input being configured to receive at least one phase of the multiple phases of second power. In at least one example, the at least one phase-selection interface is configured to switch from outputting the first selected phase of the first power to outputting a third selected phase of the first power, the third selected phase chosen from among the multiple phases of the first power. In at least one example, the at least one phase-selection interface is configured to switch from outputting the second selected phase of the second power to outputting a third selected phase of the second power, the third selected phase chosen from among the multiple phases of the second power.

In at least one example, the ATS system includes a multiphase rectifier configured to receive power from one of the first input or the second input, and a converter coupled to the multiphase rectifier and configured to receive power from the multiphase rectifier, process power received from the multiphase rectifier into processed power, and provide the processed power to the at least one controller. In at least one example, a first total number of the multiple phases of first power equals a second total number of the multiple phases of second power, and the ATS system further comprises at least one respective receptacle for each respective phase of first power, and at least one respective switching device for each respective phase of first power.

Aspects of the disclosure include a method of transferring power automatically using a switching system having a first input, a second input, at least one phase-selection interface, at least one switching device, and a receptacle, the method comprising receiving, at the first input, first power having multiple phases from a first power source, receiving, by the at least one phase-selection interface, the multiple phases of first power, selecting, by the at least one phase-selection interface, a first selected phase of first power from the multiple phases of first power, outputting, by the at least one phase-selection interface, the first selected phase of first power to the at least one switching device, receiving, at the second input, second power having multiple phases from a second power source, receiving, by the at least one phase-selection interface, the multiple phases of second power, selecting, by the at least one phase-selection interface, a second selected phase of second power from the multiple phases of second power, outputting, by the at least one phase-selection interface, the second selected phase of second power to the at least one switching device, receiving, by the at least one switching device, the first selected phase of first power and the second selected phase of second power, and providing, by the at least one switching device, one of the first selected phase of first power or the second selected phase of second power to the receptacle by coupling one of

the first input or the second input to the receptacle while decoupling the other of the first input or the second input from the receptacle.

In at least one example, the method includes providing, by the at least one switching device, the first selected phase of first power to the receptacle. In at least one example, the method includes providing, by the at least one switching device, the second selected phase of second power to the receptacle. In at least one example, the method includes decoupling, by the at least one switching device, the receptacle from the first input responsive to determining that the first power provided via the first input connection falls outside a range of acceptable voltage or current values. In at least one example, the method includes decoupling, via the at least one switching device, the receptacle from the second input responsive to determining that the second power received at the second input connection falls outside of a range of acceptable voltage or current values.

Aspects of the disclosure include at least one non-transitory computer-readable medium containing thereon instructions executable by at least one processor for controlling an automatic transfer switching system having a first input to receive multiple phases of first power, a second input to receive multiple phases of second power, at least one phase-selection interface to receive the multiple phases of first power and the multiple phases of second power, at least one switching devices, and a receptacle, the instructions instructing the at least one processor to select a first selected phase of power chosen from among the multiple phases of the first power, select a second selected phase of power chosen from among the multiple phases of the second power, control the at least one phase-selection interface to output the first selected phase of power from the multiple phases of first power received by the at least one phase-selection interface, and output the second selected phase of power from the multiple phases of second power received by the at least one phase-selection interface, and control the at least one switching device to provide power to the receptacle, wherein controlling the at least one switching device to provide power to the receptacle includes controlling the at least one switching device to couple the first input to the receptacle to provide the first selected phase of power to the receptacle when the at least one switching device decouples the second input from the receptacle, and controlling the at least one switching device to couple the second input to the receptacle to provide the second selected phase of power to the receptacle when the at least one switching device decouples the first input from the receptacle.

In at least one example, the instructions further instruct the at least one processor to switch from providing the first selected phase of power to providing the second selected phase of power by controlling the at least one switching device to decouple the first input from the receptacle and couple the second input to the receptacle. In at least one example, the instructions further instruct the at least one processor to control the at least one switching device to decouple the receptacle from the first input responsive to determining that the first power falls outside a range of acceptable voltage or current values. In at least one example, the instructions further instruct the at least one processor to control the at least one switching device to decouple the receptacle from the second input responsive to determining that the second power falls outside a range of acceptable voltage or current values.

BRIEF DESCRIPTION OF THE DRAWINGS

Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

FIG. **1** illustrates a block diagram of a power system according to an example;

FIG. **2** illustrates a block diagram of a power distribution unit according to an example;

FIG. **3** illustrates a flowchart of a process for determining the source and phase of power according to an example;

FIG. **4** illustrates a block diagram of a switching system according to an example;

FIG. **5**A illustrates a block diagram of a phase-selection interface according to an example;

FIG. **5**B illustrates a block diagram of a phase-selection interface according to another example;

FIG. **6** illustrates a block diagram of a phase-selection interface according to an example;

FIG. **7** illustrates a block diagram of a switching system according to an example;

FIG. **8**A illustrates a schematic diagram of a switching topology according to an example; and

FIG. **8**B illustrates a schematic diagram of a switching topology according to another example.

DETAILED DESCRIPTION

Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for irreconcilable differences, the term usage in this document controls.

Power sources may provide power to one or more loads. In some power-intensive applications, the loads that draw power may receive power from more than one power source. Power distribution units (PDUs) may route power from the power sources to the loads. PDUs may include automatic transfer switches that automatically switch between power sources based on a given criterion or set of criteria. In many examples, the power sources will provide alternating current (AC) power having multiple phases. However, providing multiple phases of power to multiple loads simultaneously can pose challenges. For example, ensuring that the loads are balanced across the multiple phases may be important, since imbalances may affect the quality of the power received by the loads. Furthermore, an overloaded phase of power may trip a circuit breaker. For example, a multi-pole circuit breaker conducting multi-phase power may trip on all poles (corresponding to all phases of power) if any of the poles of the circuit breaker are overloaded.

As an example, consider a data center where multiple servers are operating simultaneously and therefore drawing power simultaneously. If an AC power source provides three phases of power, but all of the servers are using only the first phase of power, power distribution may be imbalanced. Such a power imbalance may place a substantial strain on the AC power source during provision of the first phase of power, but relatively little strain during provision of the other two phases of power. Under these circumstances, the servers may not operate at a desired level of performance. In some cases, the servers may even shut down if, for example, a circuit breaker is tripped by an overloaded phase of power.

In some examples disclosed herein, a phase selection system incorporated into an automatic transfer switch or other power system may receive a multi-phase-power input and output single-phase power. Each of the single phases may be output via a respective power receptacle. The receptacles may be grouped such that, if loads are connected to the receptacles in order, power is balanced between the multiple phases. In various examples, a phase selection system may be capable of routing any of the multiple phases of the multi-phase-power input to any of the single-phase receptacles. Accordingly, examples of the disclosure provide a multiphase-to-single-phase power device capable of routing any of the multiple phases to any of the single-phase receptacles.

Continuing with the example of the data center, in some situations an AC power source may lose power or may be unable to provide acceptable power to the servers. For example, the servers may draw more power than the maximum total output of the AC power source. As a result, the servers may not be receiving acceptable power and may even shut down.

In some examples disclosed herein, a power source selection system incorporated into an automatic transfer switch or other power system may be used to select between multiple power sources to provide power to the loads. For example, if a power source cannot provide acceptable power, the power source selection system may switch some or all of the loads from the power source to another power source. Likewise, if a power source fails, the power source selection system may switch to the other power source to allow some or all of the loads to continue operation.

FIG. **1** illustrates a power system **100** according to an example. The power system **100** includes a first power source **102**, a second power source **104**, an automatic transfer switch **106** ("ATS **106**"), and an arbitrary number of n loads including a first load **108** and an nth load **110**. The ATS **106** is configured to route power from the first power source **102** and/or second power source **104** to the n loads, including the first load **108** and the nth load **110**. The ATS **106** can determine which phase of power from the first power source **102** and/or second power source **104** is provided to the n loads, including the first load **108** and the nth load **110**, and may determine which of the first power source **102** and second power source **104** provides the power to a given load. Thus, the ATS **106** can dynamically determine the phase and source of the power provided to a given load.

The first power source **102** is coupled to the ATS **106**. The second power source **104** is coupled to the ATS **106**. The first power source **102** and/or second power source **104** may include AC power sources, such as mains utility sources. The first load **108** is coupled to the ATS **106**. The nth load **110** is coupled to the ATS **106**. The first load **108** and/or nth load **110** may include power-consuming devices such as computer servers, cooling units, and so forth. One or more additional loads may also be coupled to the ATS **106**. Likewise, in some examples, the ATS **106** may be coupled to more than two power sources.

The first power source **102** is configured to provide the multiple phases of first power to the ATS **106**. The second power source **104** is configured to provide the multiple phases of second power to the ATS **106**.

The ATS **106** is configured to route power from one of the first power source **102** and second power source **104** to a given load. The ATS **106** is further configured to determine which phase of the first power and which phase of the second power is provided to a given load. For example, with respect to the first load **108**, the ATS **106** may determine whether to provide the first load **108** with power from the first power source **102** or from the second power source **104**, and may also determine which phase of the first power or the second power to provide to the first load **108**. The ATS **106** may make the same determinations with respect to every load coupled to the ATS **106**, including the nth load **110**.

The ATS **106** may be further configured to dynamically and/or selectively switch the phases of power provided to a given load or the source of the power provided to a given load. For example, with respect to the first load **108**, the ATS **106** may initially provide a first phase of first power to the first load **108**, but could change the power provided to the first load **108** to a different phase of the first power or to any phase of second power. The ATS **106** may perform the same operations with respect to every other load, including the nth load **110**.

The first load **108** and the nth load **110**, as well as any other loads coupled to the ATS **106**, may be electronic devices and/or equipment that draw power and use the power to perform functions. In some examples, the loads, including the first load **108** and the nth load **110**, may be servers, computers, or other devices capable of performing processing operations.

FIG. **2** illustrates a block diagram of an automatic transfer switch **200** ("ATS **200**") according to an example. The ATS **200** may be an example of the ATS **106** of FIG. **1**. The ATS **200** is configured to dynamically and/or selectively determine the phase and source of power provided at a given receptacle, each of which may be coupled to a respective one of the n loads. The ATS **200** includes a first input **202**, a second input **204**, a first contactor **206**, a second contactor **208**, one or more first sensors **210** ("first sensor **210**"), at least one controller **212** ("controller **212**"), one or more second sensors **214** ("second sensor **214**"), a first switching system **216**, a second switching system **218**, a third switching system **220**, a first receptacle **222**, a second receptacle **224**, a third receptacle **226**, a first rectifier **228**, a second rectifier **230**, a first converter **232**, a second converter **234**, a first circuit breaker **236**, a second circuit breaker **238**, and a third circuit breaker **240**. As illustrated, the ATS **200** is configured to receive power from three-phase power sources. Other examples may be configured to receive any number of phases of power.

The first input **202** is coupled to the first contactor **206** and to the first rectifier **228**. The first input **202** may be configured to receive three phases of first power. As a result, a first-phase first-power (that is, a first phase of first power) connection (or "first phase input") of the first input **202** may be coupled to a respective connection of the first contactor **206** and of the first rectifier **228**, a second-phase first-power connection (or "second phase input") of the first input **202** may be coupled to a respective connection of the first contactor **206** and of the first rectifier **228**, and a third-phase first-power connection (or "third phase input") of the first input **202** may be coupled to a respective connection of the first contactor **206** and of the first rectifier **228**.

The second input **204** is coupled to the second contactor **208** and to the second rectifier **230**. The second input **204** may be configured to receive three phases of second power. As a result, a first-phase second-power (that is, a first phase of second power) connection (or "fourth phase input") of the second input **204** may be coupled to a respective connection of the second contactor **208** and of the second rectifier **230**, a second-phase second-power connection (or "fifth phase input") of the second input **204** may be coupled to a respective connection of the second contactor **208** and of the second rectifier **230**, and a third-phase second-power connection (or "sixth phase input") of the second input **204** may be coupled to a respective connection of the second contactor **208** and of the second rectifier **230**.

The first contactor **206** is coupled to the first switching system **216**, the second switching system **218**, the third switching system **220**, and the first converter **232**. The first contactor **206** may have connections for providing the first, second, and third phases of first power to the switching systems. As a result, a first-phase first-power connection of the first contactor **206** may be coupled to a respective connection of the first switching system **216**, a respective connection of the second switching system **218**, and a respective connection of the third switching system **220**. A second-phase first-power connection of the first contactor **206** may be coupled to a respective connection of the first switching system **216**, a respective connection of the second switching system **218**, and a respective connection of the third switching system **220**. A third-phase first-power connection of the first contactor **206** may be coupled to a respective connection of the first switching system **216**, a respective connection of the second switching system **218**, and a respective connection of the third switching system **220**. The first contactor **206** may be energized by the first converter **232**, which dictates the switching state (for example, open and non-conducting if the first contactor **206** is not energized or closed and conducting if the first contactor **206** is energized) of the first contactor **206**.

The second contactor **208** is coupled to the first switching system **216**, the second switching system **218**, the third switching system **220**, and the second converter **234**. The second contactor **208** may have connections for providing the first, second, and third phases of second power to the switching systems. As a result, a first-phase second-power connection of the second contactor **208** may be coupled to a respective connection of the first switching system **216**, a respective connection of the second switching system **218**, and a respective connection of the third switching system **220**. A second-phase second-power connection of the second contactor **208** may be coupled to a respective connection of the first switching system **216**, a respective connection of the second switching system **218**, and a respective connection of the third switching system **220**. A third-phase second-power connection of the second contactor **208** may be coupled to a respective connection of the first switching system **216**, a respective connection of the second switching system **218**, and a respective connection of the third switching system **220**. The second contactor **208** may be energized by the second converter **234**, which dictates the switching state (for example, open and non-conducting if the second contactor **208** is not energized or closed and conducting if the second contactor **208** is energized) of the second contactor **208**.

The first sensor **210** may be coupled to one or more connections between the first contactor **206** and the first switching system **216**, the second switching system **218**, and/or the third switching system **220**. In some examples, the first sensor **210** may include a first sensor (for example, a current sensor) coupled in series between the first contactor **206** and the first switching system **216**, a second sensor coupled in series between the first contactor **206** and the second switching system **218**, and/or a third sensor coupled in series between the first contactor **206** and the third switching system **220**. In various examples, the first sensor **210** may include one or more sensors (for example, voltage sensors) coupled to at least one of a connection between the first contactor **206** and the first switching system **216**, a connection between the first contactor **206** and the second switching system **218**, and/or a connection between the first contactor **206** and the third switching system **220**. The first sensor **210** is also coupled to the controller **212**.

The second sensor **214** may be coupled to one or more connections between the second contactor **208** and the first switching system **216**, the second switching system **218**, and/or the third switching system **220**. In some examples, the second sensor **214** may include a first sensor (for example, a current sensor) coupled in series between the second contactor **208** and the first switching system **216**, a second sensor coupled in series between the second contactor **208** and the second switching system **218**, and/or a third sensor coupled in series between the second contactor **208** and the third switching system **220**. In various examples, the second sensor **214** may include one or more sensors (for example, voltage sensors) coupled to at least one of a connection between the second contactor **208** and the first switching system **216**, a connection between the second contactor **208** and the second switching system **218**, and/or a connection between the second contactor **208** and the third switching system **220**. The second sensor **214** is also coupled to the controller **212**.

The first switching system **216** is coupled to the first contactor **206** and the second contactor **208**. In some examples, the first switching system **216** has three independent connections to the first contactor **206** (one for each phase of first power), and three independent connections to the second contactor **208** (one for each phase of second power). The first switching system **216** is also coupled to the first receptacle **222** via the first circuit breaker **236**.

The second switching system **218** is coupled to the first contactor **206** and the second contactor **208**. In some examples, the second switching system **218** has three independent connections to the first contactor **206** (one for each phase of first power), and three independent connections to the second contactor **208** (one for each phase of second power). The second switching system **218** is also coupled to the second receptacle **224** via the second circuit breaker **238**.

The third switching system **220** is coupled to the first contactor **206** and the second contactor **208**. In some examples, the third switching system **220** has three independent connections to the first contactor **206** (one for each phase of first-power), and three independent connections to the second contactor **208** (one for each phase of second-power). The third switching system **220** is also coupled to the third receptacle **226** via the third circuit breaker **240**.

The first rectifier **228** is coupled to, and is configured to receive power from, the first input **202** at a plurality of input connections. The first rectifier **228** is also coupled to the first converter **232** at an output connection. The first rectifier **228** may be configured to receive one or more phases of AC power from the first input **202**, to rectify the AC power, and to provide a single rectified DC output to the first converter **232**. If power is not available from at least one phase at the first input **202**, then the first rectifier **228** may not provide power to the first converter **232**. If power is available from at least one phase at the first input **202**, then in some examples the first rectifier **228** may provide rectified output power to the first converter **232** even if one or more other phases of power are not available at the first input **202**.

The second rectifier **230** is coupled to, and is configured to receive power from, the second input **204** at a plurality of input connections. The second rectifier **230** is also coupled to the second converter **234** at an output connection. The second rectifier **230** may be configured to receive one or more phases of AC power from the second input **204**, to rectify the AC power, and to provide a single rectified DC output to the second converter **234**. If power is not available from at least one phase at the second input **204**, then the second rectifier **230** may not provide power to the second converter **234**. If power is available from at least one phase at the second input **204**, then in some examples the second rectifier **230** may provide rectified output power to the second converter **234** even if one or more other phases of power are not available at the second input **204**.

The first converter **232** is coupled to the first rectifier **228** at an input, and is coupled to the first contactor **206** and the controller **212** at respective outputs. The first converter **232** receives power from the first rectifier **228**, converts the received power, and provides converted power to the first contactor **206** and the controller **212**. For example, the first converter **232** may be a DC/DC converter configured to adjust a voltage level of the received power. The power provided from the first converter **232** to the first contactor **206** energizes the first contactor **206** and causes the first contactor **206** to close and conduct. If the first converter **232** does not energize the first contactor **206** (for example, because power is unavailable from the first input **202**), then the first contactor **206** remains open and non-conducting.

The second converter **234** is coupled to the second rectifier **230** at an input, and is coupled to the second contactor **208** and the controller **212** at respective outputs. The second converter **234** receives power from the second rectifier **230**, converts the received power, and provides converted power to the second contactor **208** and the controller **212**. For example, the second converter **234** may be a DC/DC converter configured to adjust a voltage level of the received power. The power provided from the second converter **234** to the second contactor **208** energizes the second contactor **208** and causes the second contactor **208** to close and conduct. If the second converter **234** does not energize the second contactor **208** (for example, because power is unavailable from the second input **204**), then the second contactor **208** remains open and non-conducting.

The contactors **206**, **208**, in combination with the rectifiers **228**, **230** and the converters **232**, **234**, prevent backfeed when the inputs **202**, **204** are not energized by power sources. The contactors **206**, **208** are open and non-conducting (and thus isolate the inputs **202**, **204** from the remainder of the ATS **200**) unless power is available at the inputs **202**, **204** to energize the contactors **206**, **208**. When the contactors **206**, **208** are energized, then the contactors **206**, **208** may be closed and conducting. Otherwise, if the contactors **206**, **208** are not energized (for example, because power is unavailable at the inputs **202**, **204**, respectively), then the contactors **206**, **208** are open and non-conducting. The open contactors **206**, **208** isolate the inputs **202**, **204** from the rest of the ATS **200**. This isolation may prevent backfeed to the inputs **202**, **204** from other parts of the ATS **200**.

The first input **202** is configured to receive three phases of first power. That is, the first input **202** is configured to be coupled to a first power source (for example, the first power source **102**) configured to provide three phases of first power. The first input **202** is configured to provide each phase of first power to the first contactor **206**.

The second input **204** is configured to receive three phases of second power. That is, the second input **204** is configured to be coupled to a second power source (for example, the second power source **104**) configured to provide three phases of second power. The second input **204** is configured to provide each phase of second power to the second contactor **208**.

The first contactor **206** is configured to receive the phases of first power from the first input **202**, and relay those phases of power to the first switching system **216**, the second switching system **218**, and/or the third switching system **220**. The first contactor **206** may also operate as a switching device (for example, a relay, dip-switch, limit switch, transistor-based switch, and so forth) to control whether any or all phases of first power are provided to the switching systems, including the first switching system **216**, the second switching system **218**, and/or the third switching system **220**. For example, in the event of an overcurrent condition affecting the first phase of the first power, the first contactor **206** could stop providing the first phase of the first power to the rest of a system (for example, the first switching system **216**, the second switching system **218**, and/or the third switching system **220**, and the first sensor **210**). In the event of an overcurrent condition affecting additional phases of the first power, the first contactor **206** may stop providing those additional phases of power to a system as well. In general, the first contactor **206** may stop providing any phases of the first power from the rest of the ATS **200**.

The second contactor **208** is configured to receive the phases of second power from the second input **204**, and relay those phases of power to the first switching system **216**, the second switching system **218**, and/or the third switching system **220**. The second contactor **208** may also operate as a switching device (for example, a relay, dip-switch, limit switch, transistor-based switch, and so forth) to control whether any or all phases of second power are provided to the switching systems, including the first switching system **216**, the second switching system **218**, and/or the third switching system **220**. For example, in the event of an overcurrent condition affecting the first phase of second power, the second contactor **208** could stop providing the first phase of second power to the rest of the system (for example, to the first switching system **216**, the second switching system **218**, the third switching system **220**, and/or and the second sensor **214**). In the event of an overcurrent condition affecting additional phases of second power, the second contactor **208** could stop providing those additional phases to the system as well. In general, the second contactor **208** may stop providing any phases of the second power from the rest of the ATS **200**.

The first sensor **210** is configured to sense one or more characteristics of the phases of the first power. Characteristics of power may generally include voltage, current, phase, frequency, period, harmonic content, and so forth. The first sensor **210** may include more than one sensor, and/or may include multiple sensor connections. Thus, the first sensor **210** may independently monitor the first phase of first power, the second phase of the first power, and/or the third phase of the first power, and may include multiple (for example, three) sensors to perform the monitoring.

The second sensor **214** is configured to sense one or more characteristics of the phases of the second power. Characteristics of power may generally include voltage, current, phase, frequency, period, harmonic content, and so forth. The second sensor **214** may include more than one sensor, and/or may include multiple sensor connections. Thus, the second sensor **214** may independently monitor the first phase of second power, the second phase of the second power, and/or the third phase of the second power, and may include multiple (for example, three) sensors to perform the monitoring.

In various examples, the first switching system **216**, the second switching system **218**, and/or the third switching system **220** are each configured to select between phases of power and sources of power. Each of the first switching system **216**, the second switching system **218**, and/or the third switching system **220** is configured to provide output power to a respective receptacle.

The first switching system **216** is configured to provide first output power to the first receptacle **222** via the first circuit breaker **236**. The second switching system **218** is configured to provide second output power to the second receptacle **224** via the second circuit breaker **238**. The third switching system **220** is configured to provide third output power to the third receptacle **226** via the third circuit breaker **240**. The first output power, second output power, and third output power may each be single-phase power. Each of the circuit breakers **236-240** may trip (that is, transition from being closed and conducting to being open and non-conducting) if a current conducted by the respective circuit breaker exceeds a current rating. For example, if the current through the first circuit breaker **236** exceeds the current rating of the first circuit breaker **236**, then the first circuit breaker **236** may trip and thereby disconnect the first switching system **216** from the first receptacle **222**.

Each of the first switching system **216**, the second switching system **218**, and/or the third switching system **220** may be configured to provide a single phase of first power or a single phase of second power to a respective one of the first receptacle **222**, second receptacle **224**, and/or third receptacle **226**. The first switching system **216** may be configured to provide the first phase of the first power or the first phase of the second power, the second phase of the first power or the second phase of the second power, or the third phase of the first power or the third phase of the second power to the first receptacle **222**. The second switching system **218** may be configured to provide the first phase of the first power or the first phase of the second power, the second phase of the first power or the second phase of the second power, or the third phase of the first power or the third phase of the second power to the second receptacle **224**. The third switching system **220** may be configured to provide the first phase of the first power or the first phase of the second power, the second phase of the first power or the second phase of the second power, or the third phase of the first power or the third phase of the second power to the third receptacle **226**. More generally, each of the first switching system **216**, the second switching system **218**, and/or the third switching system **220** may freely and independently output any given phase of the first power or the second power. Accordingly, each of the first receptacle **222**, second receptacle **224**, and/or third receptacle **226** may receive output power derived from a different phase or different source compared to any other receptacle.

In at least one example, however, each of the first switching system **216**, the second switching system **218**, and/or the third switching system **220** may output, to a respective one of the first receptacle **222**, second receptacle **224**, and/or third receptacle **226**, power from the same source but with a different phase. For example, the first switching system **216** may provide a first phase of the first power to the first receptacle **222**; the second switching system **218** may provide a second phase of the first power to the second receptacle **224**; and the third switching system **220** may provide a third phase of the first power to the third receptacle **226**.

In another example, the first switching system **216** may provide a first phase of the second power to the first receptacle **222**; the second switching system **218** may provide a second phase of the second power to the second receptacle **224**; and the third switching system **220** may provide a third phase of the second power to the third receptacle **226**.

In yet another example, the first switching system **216** may provide a second phase of the first power to the first receptacle **222**; the second switching system **218** may provide a third phase of the first power to the second receptacle **224**; and the third switching system **220** may provide a first phase of the first power to the third receptacle **226**. The first switching system **216**, the second switching system **218**, and/or the third switching system **220** will be discussed in further detail with respect to FIG. **4**.

The first receptacle **222** is configured to provide first output power to a first load connected to the first receptacle **222** (for example, the first load **108**). The second receptacle **224** is configured to provide second output power to a second load. The third receptacle **226** is configured to provide third output power to a third load. The receptacles may include, for example, electrical outlets configured to provide power to a connected load. The controller **212** is configured to control the first switching system **216**, the second switching system **218**, and/or the third switching system **220**, the first contactor **206** and/or second contactor **208**, and/or the first sensor **210** and/or second sensor **214**. The controller **212** may also be configured to communicate with loads and/or power sources (for example, the first power source **102**, second power source **104**, and/or any of the n loads including the first load **108** and/or nth load **110**). The controller **212** may determine whether the first contactor **206** and/or second contactor **208** should be in an open state or a closed state and control the first contactor **206** and/or second contactor **208** accordingly. The controller **212** may further determine a desired power source and power phase to provide to the first receptacle **222**, second receptacle **224**, and/or third receptacle **226**, and may control the first switching system **216**, the second switching system **218**, and/or the third switching system **220** accordingly.

The ATS **200** generally operates as follows. Power may be received at the inputs **202**, **204** from the power sources **102**, **104**, respectively. If power is available at the inputs **202**, **204**, then the contactors **206**, **208** are energized by the power received at the inputs **202**, **204**. The contactors **206**, **208** close in response to being energized and provide power to the switching systems **216-220**. If power is not available at one of the inputs **202**, **204**, then the corresponding one of the contactors **206**, **208**, respectively, is not energized and remains open. For example, if acceptable power is not available at the first input **202**, then the first contactor **206** remains open.

Each of the switching systems **216-220** receives power from one or both of the contactors **206**, **208**. Each switching system **216-220** selects one of the inputs **202**, **204** to draw power from. If power is only available from one of the inputs **202**, **204**, then the switching systems **216-220** draw power only from that input. If power is available from both of the inputs **202**, **204**, then the switching systems **216-220** may default to drawing power from a preferred one of the inputs **202**, **204**. Either of the inputs **202**, **204** may be arbitrarily selected as a preferred input. For example, the first input **202** may arbitrarily be selected as a preferred input. In this example, the switching systems **216-220** preferably draw power from the first input **202** if power is available from both of the inputs **202**, **204**. The controller **212** may control the switching systems **216-220** to switch to drawing power from the second input **204** if acceptable power is no longer available at the first input **202**.

The selected power source may provide multiple phases of power. The controller **212** may decide which phase of power is to be provided to the receptacles **222-226**. For example, if the first input **202** provides three-phase power with phases L1*a*, L2*a*, and L3*a*, then the controller **212** may control each of the switching systems **216-220** to provide a selected one of these three phases to a respective receptacle. Each of the switching systems **216-220** may provide the same or a different phase of power to a respective one of the receptacles **222-226**. For example, the first switching system **216** may provide phase L1*a* to the first receptacle **222**, the second switching system **218** may provide phase L2*a* to the second receptacle **224**, and the third switching system **220** may provide phase L3*a* to the third receptacle **226**.

Accordingly, the controller **212** may control the switching systems **216-220** to route any phase of power from any power source to any one of the receptacles **222-226**. The controller **212** may control the switching systems **216-220** to switch power sources if, for example, a selected power source is no longer able to provide acceptable power. The controller **212** may control the switching systems **216-220** to switch which phase of power is provided to the receptacles **222-226** to, for example, balance the load across the phases. The ATS **200** may therefore provide redundant three-to-single-phase power while dynamically balancing power drawn across multiple phases.

FIG. **3** illustrates a process **300** for determining the source and phase of power to provide to a load according to an example. For example, the loads may include loads coupled to the receptacles **222-226**. Over the course of operation of the ATS **200**, the source of the power may be switched from one power supply to another. For example, the source of power may be switched from the first input **202** (which may be coupled to the first power source **102**) to the second input **204** (which may be coupled to the second power source **104**). Likewise, the phase of power provided to the load may be adjusted, for example, to maintain a balanced distribution of loads across numerous phases of power. For example, the phase of power may be adjusted from a first of three phases of three-phase power received at the first input **202** to a second of three phases of three-phase power received at the first input **202**.

For purposes of example, the process **300** is described as beginning after the controller **212** has determined which phases of power to provide to respective loads, and the ATS **200** is providing those respective phases of power to the respective loads. As discussed below, the controller **212** may thus have controlled or be controlling the switching systems **216-220** to establish desired source-to-load conductive paths from the first input **202** and/or the second input **204** to the receptacles **222-226**.

At act **302**, the inputs **202**, **204** receive power from the power sources **102**, **104**, respectively. The contactors **206**, **208** may close responsive to being energized by the power sources **102**, **104**. Power is therefore provided from the inputs **202**, **204** to the switching systems **216-220** via the contactors **206**, **208**. The controller **212** monitors power characteristics of the power provided by the first power source **102** and/or the second power source **104** to the receptacles **222-226**. The controller **212** may receive power-characteristic information indicative of the power characteristics (for example, a voltage, a current, and so forth) from the sensors **210**, **214**. Because act **302** is executed after the controller **212** has previously determined which phases of power to provide to which of the receptacles **222-226**, the controller **212** may be capable of correlating the power-characteristic information to each source-to-receptacle connection. For example, the controller **212** may be capable of determining the source and phase of the power provided to each of the receptacles **222-226** and may have access to power-characteristic information for each. The process **300** may continue to act **304**.

At act **304**, the controller **212** determines whether to change the power source providing power to one or more of the loads. For example, the controller **212** may determine whether the power quality provided to at least one load ("the loads") is acceptable for the load. The controller **212** may determine whether the power quality is acceptable for the loads based on one or more criteria. For example, the controller **212** may determine whether the characteristics of the power (for example, voltage, current, phase, frequency, period, harmonic content, and so forth) determined at act **302** fall within a certain range or ranges of values. As an example, suppose the loads requires power provided at 50 A of current, with a tolerance of +/−1 A. The power quality may be acceptable if the current of the power is within a range of 49 to 51 A, but may not be acceptable if the power is provided outside that range. Characteristics may be required (to be considered acceptable) to fall within a given range, or to be above or below a threshold value.

The controller **212** may determine, in addition or alternatively to determining whether the characteristics of the power are acceptable, whether one power source is under undue strain. Undue strain may include the difference in power provided by one power source compared to another exceeding a threshold percentage or magnitude. For example, a power source (such as the first power source **102**) may be providing power to a larger number of the loads and/or providing power to loads that consume substantially more power as compared to another power source (such as the second power source **104**). Act **304** may therefore include the controller **212** determining whether the difference in power provided by the power sources **102**, **104** exceeds a threshold amount (such as a threshold amount of power or a threshold difference in power) or falls outside a target range).

In various examples, the controller **212** may determine whether the power provided by a power source is acceptable based on the characteristics of the power, the distribution of power consumption between the power sources (such as the power sources **102**, **104**), the number of loads drawing power from the power sources, and/or so forth. That is, the controller **212** may transfer a given load to a given power source based on the power quality, and/or to more evenly distribute the number of loads across the power sources, and/or to more evenly distribute the power consumption of loads across the power sources, and so forth. In various examples, the controller **212** may not switch every load of the loads to a new power source. In some examples, the controller **212** may allow some of the one or more loads of the loads to continue receiving power from the original power source, for example, because the power quality improves, the power sources become balanced, or the strain is alleviated (for example, by the transfer of other loads to the other power sources).

If the controller **212** determines that the power quality is acceptable (**304** YES), the process **300** may continue to act **308**. If the controller **212** determines that the power quality is not acceptable (**304** NO), the process **300** may continue to act **306**.

At act **306**, the controller **212** may control a respective switching system (for example, the switching systems **216-220**) to change a power source that is providing power to one or more respective loads. The controller **212** may transfer the one or more loads from a first power source (for example, the first power source **102**) to a second power source (for example, the second power source **104**). The controller **212** may control a switching system (for example, the first switching system **216**) to change from one power source to another by controlling a phase-selection interface of the switching system to switch from drawing power from a power source and/or input to a different power source and/or input. The process **300** may return to act **302**.

The process **300** may then return to act **304**. If the controller **212** determines at act **304** that a power source is not to be changed (**304** YES), then the process **300** continues to act **308**.

At act **308**, the controller **212** determines whether to change the phase of power provided to the loads. The controller **212** may determine whether the one or more phases of the power provided to the loads are acceptable for the load, and may determine to change the phase of power provided to a given load based on that phase of power being acceptable or not. As with act **304**, numerous parameters may be employed to make this determination. The controller **212** may determine whether to change the phase of power provided to one or more loads of the loads based on one or more phase-change criteria.

For example, the controller **212** may determine that the characteristics of the power fall outside a desired range or exceed (or fall below) a given threshold (for example, a current, voltage frequency, or voltage magnitude threshold and/or range). Such a determination may include, for example, the controller **212** determining whether the voltage, frequency, and/or current fall outside of an acceptable range. The controller **212** may also determine whether a given phase of power is under undue strain. A given phase of power may be under undue strain if the difference in power provided by that phase of power compared to another phase of power (for example, from the same power source) exceeds a threshold amount or falls outside of a desired range.

If the controller **212** determines not to change the phase of power (**308** NO), the process **300** may continue to act **302**. In some examples, returning to act **302** from act **308** may indicate that—at least for the time being—the power source and the phases of power are providing power effectively, such as by having acceptable power characteristics, being evenly distributed, and/or not being under undue strain.

Otherwise, if the controller **212** determines that the phase of power should be changed (**308** YES), then the process **300** continues to act **310**. In some examples, continuing to act **310** may indicate that it may be desirable to change the phase of power provided to at least one load because, for example, power is not evenly distributed between the phases of a power source. The controller **212** may not switch every load of the loads to a new phase of power. In some examples, the controller **212** may allow some of the one or more loads of the loads to continue receiving the original phase of power, for example, because the power quality improves, the phases become balanced, or the strain is alleviated (for example, by the transfer of other loads to the other phases of power). The controller **212** may determine which phase of power to switch to based on the characteristics of that phase of power, the distribution of power consumption between loads, the number of loads drawing power from a given phase of power, and/or so forth. That is, the controller **212** may transfer a given load to a given phase of power based on the power quality, and/or to more evenly distribute the number of loads across the phases of power, and/or to more evenly distribute the power consumption of loads across the phases of power (that is, reduce undue strain), and/or so forth.

At act **310**, the controller **212** controls a phase-selection interface of a switching device to transition one or more loads of the loads from a first phase of power to a different phase of power provided by the same power source as provided the first phase of power. For example, the controller **212** may control a phase-selection interface to switch from providing a first phase of power to providing a second phase of power. The process **300** may continue to act **302**.

FIG. **4** illustrates a switching system **400** coupled to a receptacle **414** according to an example. The switching system **400** may be one example of an implementation of any of the switching systems **216-220**, and the receptacle **414** may be an example of any of the receptacles **222-226**. A controller, such as the controller **212**, may control the switching system **400** to dynamically and selectively change which power source and which power phase is provided to a load coupled to the receptacle **414**, thus allowing the switching system **400** to control the exact phase and source of the power provided at to the receptacle **414**.

The switching system **400** includes a phase-selection interface **402** and a switching device **408**. The phase-selection interface **402** includes a first phase selector **404** and a second phase selector **406**. The switching device **408** includes a first switch **410** and a second switch **412**. The first switch **410** includes a first power switch **410*a*** and a first ground switch **410*b***. The second switch **412** includes a second power switch **412*a*** and a second ground switch **412*b***. The switching system **400** also includes a reference-voltage connection **416** ("reference connection **416**"), which may be coupled to a reference-voltage node, such as a neutral node.

The phase-selection interface **402** is coupled to the switching device **408**. The switching device **408** is coupled to the phase-selection interface **402** and is coupled to the receptacle **414**. The reference connection **416** is coupled to the switching device **408**. In more detail, the first phase selector **404** is coupled to the switching device **408** at a first terminal of the first power switch **410***a* of the first switch **410**. The second phase selector **406** is coupled to the switching device **408** at a first terminal of the second power switch **412***a* of the second switch **412**. The reference connection **416** is coupled to the switching device **408** at a first terminal of the first ground switch **410***b* of the first switch **410** and at a first terminal of the second ground switch **412***b* of the second switch **412**. The receptacle **414** may have a first connection and a second connection. The first connection of the receptacle **414** may be coupled to the switching device **408** at a second terminal of the first power switch **410***a* of the first switch **410** and at a second terminal of the second power switch **412***a* of the second switch **412**. The second connection of the receptacle **414** may be coupled to the switching device **408** at a second terminal of the first ground switch **410***b* of the first switch **410** and at a second terminal of the second ground switch **412***b* of the second switch **412**.

The phase-selection interface **402** is configured to receive one or more phases of power from the first power source **102** and one or more phases of power from the second power source **104**. In some examples, the phase-selection interface **402** may receive all available phases of power from both power sources **102**, **104**. In FIG. **4**, the phases of power received from the first power source **102** are labeled A, B, and C, and the phases of power received from the second power source **104** are labeled D, E, and F.

Within the phase-selection interface **402**, the first phase selector **404** is configured to receive the A, B, and C phases of first power from the first power source **102**, and the second phase selector **406** is configured to receive the D, E, and F phases of second power from the second power source **104**. The first phase selector **404** has a single output (connected to the switching device **408** at the first terminal of the first power switch **410***a*). The first phase selector **404** is configured to route one of the A, B, and C phases of first power to the switching device **408** (for example, at the first power switch **410***a*) via the output. The second phase selector **406** also has a single output (connected to the switching device **408** at the first terminal of the second power switch **412***a*). The second phase selector **406** is configured to route one of the D, E, and F phases of the second power to the switching device **408** (for example, to the second power switch **412***a*) via the output. As discussed in greater detail below with respect to FIGS. **8**A and **8**B, the first phase selector **404** and the second phase selector **406** may have switching, routing, or other control systems internally to selectively route a given phase of power to the switching device **408** while blocking the other phases of power.

The switching device **408** is configured to select between power derived from the first power source **102** and power derived from the second power source **104** to provide to the receptacle **414**. In some examples, the switching device **408** may receive a phase of the first power and a phase of the second power simultaneously, and may route one of those phases to the receptacle **414**. For example, if the switching device **408** receives the A phase of power from the first phase selector **404** and the D phase of power from the second phase selector **406**, the switching device **408** may route either the A phase of power or the D phase of power to the receptacle **414**.

Within the switching device **408**, the first switch **410** controls the connection between the first power source **102** and the receptacle **414**, and the second switch **412** control the connection between the second power source **104** and the receptacle **414**. In particular, the first power switch **410***a* controls whether the output of the first phase selector **404** is coupled to the receptacle **414**. When the first power switch **410***a* is closed and conducting, the output of the first phase selector **404** is coupled to the receptacle **414**. When the first power switch **410***a* is open and non-conducting, the output of the first phase selector **404** is decoupled from the receptacle **414**. The second power switch **412***a* controls when the output of the second phase selector **406** is coupled to the receptacle **4143**. When the second power switch **412***a* is closed and conducting, the output of the second phase selector **406** is coupled to the receptacle **414**. When the second power switch **412***a* is open and non-conducting, the output of the second phase selector **406** is decoupled from the receptacle **414**.

Both the first switch **410** and the second switch **412** contain respective reference-node switches (the first reference-node switch **410***b* and the second reference-node switch **412***b*). When either reference-node switch is closed, the receptacle **414** is coupled to the reference connection **416**.

The first power switch **410***a* and first ground switch **410***b* may be operated in tandem, such that when one is closed the other is also closed, and when one is open the other is also open. Likewise, the second power switch **412***a* and second ground switch **412***b* may be operated in tandem, such that when one is closed the other is also closed, and when one is open the other is also open. In some examples, the first switch **410** and second switch **412** may be operated in complementary fashion such that when one is open (and not conducting) the other is closed (and conducting). For example, when the first switch **410** is open the second switch **412** is closed, and when the first switch **410** is closed the second switch **412** is open. In some examples, the first switch **410** and second switch **412** may both be open at the same time.

FIG. **5**A illustrates a phase-selection interface **502** according to an example. As will be apparent by comparing the phase-selection interface **502** to the phase-selection interface **402** of FIG. **4**, the phase-selection interface **502** has only a single phase selector **504**. The phase selector **504** is configured to receive each of the A, B, C, D, E, and F phases of the first power source **102** and the second power source **104**. The phase selector **504** is configured to provide one phase of power from among the A, B, C, D, E, and F phases of power at a first output (for example, the "X" output) and one phase of the A, B, C, D, E, and F phases of power at the second output (for example, the "Y" output). In some examples, the A, B, and C phases of power may be provided by a first power source, and the D, E, and F phases of power may be provided by a second power source.

FIG. **5**B illustrates a phase-selection interface **552** according to another example. As will be apparent by comparing the phase-selection interface **552** to the phase-selection interface **402** of FIG. **4**, the phase-selection interface **522** has one phase selector for each phase of power (including a first phase selector **554**, a second phase selector **556**, and a third phase selector **558**). Each of the phase selectors **554**-**558** is configured to receive a single phase of first power or a single phase of second power. For example, the first phase selector **554** is configured to receive the A-phase of the first power, the second phase selector **556** is configured to receive the B-phase of the first power, and the third phase selector **558** is configured to receive the F-phase of the second power. Each phase selector of the phase-selection interface **552** may control whether the phase of received power is available to an output (the "X" and/or "Y" outputs) of the phase-selection interface **552**. In some examples, one of the A, B, or C phase of first power may be provided at the X output, and one of the D, E, or F phases of second power may be provided at the Y output.

Accordingly, various implementations of the phase-selection interface **402** are within the scope of the disclosure. The phase-selection interface **402** may include the phase selectors **404** and **406** of FIG. **4**, the phase selector **504** of FIG. **5**A, the phase selectors **554-558** of FIG. **5**B, or may include some other configuration of phase selectors.

FIG. **6** illustrates a block diagram of a phase selection topology **600** according to an example. The topology **600** includes a phase-selection interface **602**, a first switching device **604**, a second switching device **606**, a third switching device **608**, a first receptacle **610**, a second receptacle **612**, and a third receptacle **614**. In FIG. **6**, the phase selection for all of the switching devices **604-608** is handled at the phase-selection interface **602**, which provides a respective phase of first power and a respective phase of second power to each of the switching devices **604-608**.

The phase-selection interface **602** is coupled to the first switching device **604** via a first connection, is coupled to the second switching device **606** via a second connection, and is coupled to the third switching device **608** via a third connection. The first switching device **604** is coupled to the phase-selection interface **602** at a first connection and is coupled to the first receptacle **610** at a second connection. The second switching device **606** is coupled to the phase-selection interface **602** at a first connection and is coupled to the second receptacle **612** at a second connection. The third switching device **608** is coupled to the phase-selection interface **602** at a first connection and is coupled to the third receptacle **614** at a second connection.

The phase-selection interface **602** is configured to receive each phase of first power and each phase of second power. The phase-selection interface **602** then provides a first phase of first power and a first phase of second power to the first switching device **604**, a second phase of first power and a second phase of second power to the second switching device **606**, and a third phase of first power and a third phase of second power to the third switching device **608**. The first, second, and third phases of first power and second power may be the same phase or different phases.

The first switching device **604** routes the first phase of first power or the first phase of second power to the first receptacle **610**. The second switching device **606** routes the second phase of first power or the second phase of the second power to the second receptacle **612**. The third switching device **608** routes the third phase of first power or the third phase of second power to the third receptacle **614**.

For example, the phase-selection interface **602** may provide both the A phase of first power and the D phase of second power to the first switching device **604**. The first switching device **604** may then provide one of either the A phase of first power or the D phase of second power to the first receptacle **610**.

The first receptacle **610** provides the phase of power received from the first switching device **604** to a load. The second receptacle **612** provides the phase of power received from the second switching device **606** to a load. The third receptacle **614** provides the phase of power received from the third switching device **608** to a load.

FIG. **7** illustrates a block diagram of a plurality of switching systems **700** ("switching system **700**") according to an example. The switching system **700** may be one example of an implementation of any and/or all of the switching systems **216-220**.

The switching system **700** includes a first switching device **702**, a second switching device **704**, a third switching device **706**, a phase-selection interface **708**, a first receptacle **716**, a second receptacle **718**, and a third receptacle **720**. The phase-selection interface **708** includes a first phase selector **710**, a second phase selector **712**, and a third phase selector **714**.

The first switching device **702** is coupled to the first phase selector **710**. In some examples, the first switching device **702** may be coupled to the first phase selector **710** via a first connection, a second connection, and a third connection, where each connection carries a different phase of either the first power or the second power.

The second switching device **704** is coupled to the second phase selector **712**. In some examples, the second switching device **704** may be coupled to the second phase selector **712** via a first connection, a second connection, and a third connection, where each connection carries a different phase of either the first power or the second power.

The third switching device **706** is coupled to the third phase selector **714**. In some examples, the third switching device **706** may be coupled to the third phase selector **714** via a first connection, a second connection, and a third connection, where each connection carries a different phase of either the first power or the second power.

The first phase selector **710** is further coupled to the first receptacle **716**. The second phase selector **712** is further coupled to the second receptacle **718**, and the third phase selector **714** is further coupled to the third receptacle **720**.

The first switching device **702** is configured to receive each phase of the first power (phases A, B, and C) and each phase of the second power (phases D, E, and F). The first switching device **702** selects between the phases of the first power and the phases of the second power, and provides either the phases A, B, and C of the first power to the first phase selector **710**, or provides the phases D, E, and F of the second power to the first phase selector **710**.

The second switching device **704** is configured to receive each phase of the first power (phases A, B, and C) and each phase of the second power (phases D, E, and F). The second switching device **704** selects between the phases of the first power and the phases of the second power, and provides either the phases A, B, and C of the first power to the second phase selector **712**, or provides the phases D, E, and F of the second power to the second phase selector **712**.

The third switching device **706** is configured to receive each phase of the first power (phases A, B, and C) and each phase of the second power (phases D, E, and F). The third switching device **706** selects between the phases of the first power and the phases of the second power, and provides either the phases A, B, and C of the first power to the third phase selector **714**, or provides the phases D, E, and F of the second power to the third phase selector **714**.

The first phase selector **710** selects one phase from among the phases provided to the first phase selector **710** and provides the chosen phase to the first receptacle **716**. The second phase selector **712** selects one phase from among the phases provided to the second phase selector **712** and provides the chosen phase to the second receptacle **718**. The third phase selector **714** selects one phase from among the phases provided to the third phase selector **714** and provides the chosen phase to the third receptacle **720**.

Accordingly, the switching system **700** may be similar to the switching system **400**. For example, each of the switching systems **400**, **700** may enable a desired phase of power from a desired power source to be provided to a respective receptacle. However, whereas in FIG. **4** the receptacle **414** is coupled to the switching device **408**, in FIG. **7** the first receptacle **716**, for example, is coupled to the first phase selector **710** of the phase-selection interface **708**. That is, whereas in FIG. **4** the phase-selection interface **402** is upstream of the switching device **408**, in FIG. **7** the first switching device **702** is upstream of the phase-selection interface **708**. Accordingly, various examples of the switching systems **216-220** are within the scope of the disclosure, and an order of components may be rearranged between different examples.

FIG. **8**A illustrates a schematic diagram of a first switching topology **800** according to an example. The first switching topology **800** may represent an implementation of any of various phase-selection interfaces, phase selectors, and/or switching devices discussed herein. For example, the first switching topology **800** may represent an example implementation of either of the phase selectors **404**, **406**, or any of the phase selectors **710-714**.

The first switching topology **800** includes a first phase-selection switch **802** ("first switch **802**"), a second phase-selection switch **804** ("second switch **804**"), a third phase-selection switch **806** ("third switch **806**"), and an output **808**. Each of the phase-selection switches **802-806** may be communicatively coupled to a controller, such as the controller **212**. The first switch **802** may be configured to receive a first phase of power, the second switch **804** may be configured to receive a second phase of power, and the third switch **806** may be configured to receive a third phase of power. When the first switch **802** is closed, the first phase of power is provided to the output **808** via the conducting path created by the closed state of the first switch **802**. Likewise, when the second switch **804** is closed, the second phase of power is provided to the output **808**. When the third switch **806** is closed, the third phase of power is provided to the output **808**. When any of the switches **802-806** is open, the respective switch(es) does not provide power to the output **808**. Each switch may be opened or closed independently. However, in some examples, when one switch is closed the other two switches are open. For example, if the first switch **802** is closed (such that the first phase of power is provided to the output **808**), the second switch **804** and the third switch **806** may be open.

FIG. **8**B illustrates a schematic diagram of a second phase-selection switch topology **850** ("second switching topology **850**") according to another example. The second switching topology **850** may represent an implementation of any of various phase-selection interfaces, phase selectors, and/or switching devices discussed herein. For example, the second switching topology **850** may represent an example implementation of either of the phase selectors **404**, **406**, or any of the phase selectors **710-714**.

The second switching phase-selection switch topology **850** is an example of a single-pole triple-throw switch. The second switching topology **850** includes a first terminal **852**, a second terminal **854**, a third terminal **856**, and a pole **858**. The first terminal **852** may be configured to receive a first phase of power, the second terminal **854** may be configured to receive a second phase of power, and the third terminal **856** may be configured to receive a third phase of power. The second switching topology **850** may be coupled to a controller (for example, the controller **212**) configured to control a switching state of the second switching topology **850**. For example, the controller **212** may control the second switching topology **850** such that the pole **858** is coupled to one of the terminals **852-856**. The second switching topology **850** may therefore perform a similar function as the first switching topology **800**, albeit with a different structure.

Various controllers, such as the controller **212**, may execute various operations discussed above. Using data stored in associated memory and/or storage, the controller **212** also executes one or more instructions stored on one or more non-transitory computer-readable media, which the controller **212** may include and/or be coupled to, that may result in manipulated data. In some examples, the controller **212** may include one or more processors or other types of controllers. In one example, the controller **212** is or includes at least one processor. In another example, the controller **212** performs at least a portion of the operations discussed above using an application-specific integrated circuit tailored to perform particular operations in addition to, or in lieu of, a general-purpose processor. As illustrated by these examples, examples in accordance with the present disclosure may perform the operations described herein using many specific combinations of hardware and software and the disclosure is not limited to any particular combination of hardware and software components. Examples of the disclosure may include a computer-program product configured to execute methods, processes, and/or operations discussed above. The computer-program product may be, or include, one or more controllers and/or processors configured to execute instructions to perform methods, processes, and/or operations discussed above.

Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of, and within the spirit and scope of, this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

What is claimed is:

1. An automatic transfer switching (ATS) system comprising:

a first input configured to receive multiple phases of first power from a first power source;

a second input configured to receive multiple phases of second power from a second power source;

a receptacle configured to provide power to one or more loads;

at least one phase-selection interface configured to:

receive the multiple phases of first power from the first power source and selectively output a first selected phase of the first power chosen from among the multiple phases of the first power, and

receive the multiple phases of second power from the second power source and selectively output a second selected phase of the second power chosen from among the multiple phases of the second power;

at least one switching device coupled to the receptacle and to the at least one phase-selection interface; and

at least one controller configured to control the at least one switching device to provide power to the receptacle, wherein controlling the at least one switching device to provide power to the receptacle includes:

controlling the at least one switching device to couple the first input to the receptacle via the at least one phase-selection interface to provide the first selected phase to the receptacle when the at least one switching device decouples the second input from the receptacle, and controlling the at least one switching device to couple the second input to the receptacle via the at least one phase-selection interface to provide the second selected phase to the receptacle when the at least one switching device decouples the first input from the receptacle.

2. The ATS system of claim 1 wherein the at least one switching device includes a first switch and a second switch, the first switch being configured to selectively couple the receptacle to the first input and the second switch being configured to selectively couple the receptacle to the second input.

3. The ATS system of claim 2 wherein:

the first switch is in a closed state when the first switch couples the first input to the receptacle;

the second switch is in a closed state when the second switch couples the second input to the receptacle;

the first switch is in an open state when the first switch decouples the first input from the receptacle; and

the second switch is in an open state when the second switch decouples the second input from the receptacle.

4. The ATS system of claim 1 wherein the at least one phase-selection interface includes:

a phase selection switch configured to selectively couple at least one phase input of the first input to the receptacle, the at least one phase input being configured to receive at least one phase of the multiple phases of first power from the first power source.

5. The ATS system of claim 1 wherein the at least one phase-selection interface includes:

a phase selection switch configured to selectively couple at least one phase input of the first input to the at least one switching device, the at least one phase input being configured to receive at least one phase of the multiple phases of first power from the first power source.

6. The ATS system of claim 1 wherein the at least one phase-selection interface includes:

a phase selection switch configured to selectively couple at least one phase input of the second input to the receptacle, the at least one phase input being configured to receive at least one phase of the multiple phases of second power from the second power source.

7. The ATS system of claim 1 wherein the at least one phase-selection interface includes:

a phase selection switch configured to selectively couple at least one phase input of the second input to the at least one switching device, the at least one phase input being configured to receive at least one phase of the multiple phases of second power.

8. The ATS system of claim 1 wherein the at least one phase-selection interface is configured to switch from outputting the first selected phase of the first power to outputting a third selected phase of the first power, the third selected phase chosen from among the multiple phases of the first power.

9. The ATS system of claim 1 wherein the at least one phase-selection interface is configured to switch from outputting the second selected phase of the second power to outputting a third selected phase of the second power, the third selected phase chosen from among the multiple phases of the second power.

10. The ATS system of claim 1 further comprising:

a multiphase rectifier configured to receive power from one of the first input or the second input; and

a converter coupled to the multiphase rectifier and configured to receive power from the multiphase rectifier, process power received from the multiphase rectifier into processed power, and provide the processed power to the at least one controller.

11. The ATS system of claim 1 wherein a first total number of the multiple phases of first power equals a second total number of the multiple phases of second power, and the ATS system further comprises:

at least one respective receptacle for each respective phase of the first power; and

at least one respective switching device for each respective phase of the first power.

12. A method of transferring power automatically using a switching system having a first input, a second input, at least one phase-selection interface, at least one switching device, and a receptacle, the method comprising:

receiving, at the first input, first power having multiple phases from a first power source;

receiving, by the at least one phase-selection interface, the multiple phases of first power;

selecting, by the at least one phase-selection interface, a first selected phase of first power from the multiple phases of first power;

outputting, by the at least one phase-selection interface, the first selected phase of first power to the at least one switching device;

receiving, at the second input, second power having multiple phases from a second power source;

receiving, by the at least one phase-selection interface, the multiple phases of second power;

selecting, by the at least one phase-selection interface, a second selected phase of second power from the multiple phases of second power;

outputting, by the at least one phase-selection interface, the second selected phase of second power to the at least one switching device;

receiving, by the at least one switching device, the first selected phase of first power and the second selected phase of second power; and

providing, by the at least one switching device, one of the first selected phase of first power or the second selected phase of second power to the receptacle by coupling one of the first input or the second input to the receptacle while decoupling the other of the first input or the second input from the receptacle.

13. The method of claim 12 further comprising providing, by the at least one switching device, the first selected phase of first power to the receptacle.

14. The method of claim 12 further comprising providing, by the at least one switching device, the second selected phase of second power to the receptacle.

15. The method of claim 12 further comprising:

decoupling, by the at least one switching device, the receptacle from the first input responsive to determining that the first power provided via the first input connection falls outside a range of acceptable voltage or current values.

16. The method of claim 12 further comprising:

decoupling, via the at least one switching device, the receptacle from the second input responsive to determining that the second power received at the second input connection falls outside of a range of acceptable voltage or current values.

17. At least one non-transitory computer-readable medium containing thereon instructions executable by at least one processor for controlling an automatic transfer switching system having a first input to receive multiple phases of first power, a second input to receive multiple phases of second power, at least one phase-selection inter-

face to receive the multiple phases of first power and the multiple phases of second power, at least one switching device, and a receptacle, the instructions instructing the at least one processor to:

select a first selected phase of power chosen from among the multiple phases of the first power;

select a second selected phase of power chosen from among the multiple phases of the second power;

control the at least one phase-selection interface to

output the first selected phase of power from the multiple phases of first power received by the at least one phase-selection interface, and

output the second selected phase of power from the multiple phases of second power received by the at least one phase-selection interface; and

control the at least one switching device to provide power to the receptacle, wherein controlling the at least one switching device to provide power to the receptacle includes

controlling the at least one switching device to couple the first input to the receptacle to provide the first selected phase of power to the receptacle when the at least one switching device decouples the second input from the receptacle, and

controlling the at least one switching device to couple the second input to the receptacle to provide the second selected phase of power to the receptacle when the at least one switching device decouples the first input from the receptacle.

18. The at least one non-transitory computer-readable medium of claim **17** wherein the instructions further instruct the at least one processor to switch from providing the first selected phase of power to providing the second selected phase of power by controlling the at least one switching device to decouple the first input from the receptacle and couple the second input to the receptacle.

19. The at least one non-transitory computer-readable medium of claim **17** wherein the instructions further instruct the at least one processor to control the at least one switching device to decouple the receptacle from the first input responsive to determining that the first power falls outside a range of acceptable voltage or current values.

20. The at least one non-transitory computer-readable medium of claim **17** wherein the instructions further instruct the at least one processor to control the at least one switching device to decouple the receptacle from the second input responsive to determining that the second power falls outside a range of acceptable voltage or current values.

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